

Refining metals

THE CHEMICAL PROCESSES THAT EXTRACT PURE METALS FROM ORES.

SEE ALSO

◀ 124–125 Transition metals

◀ 129 Types of reaction

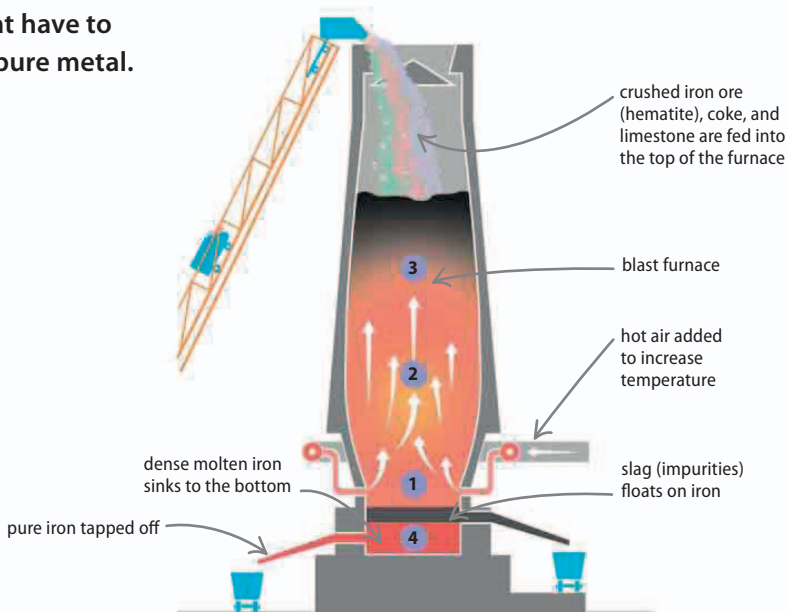
◀ 132–133 Redox reactions

◀ 148–149 Electrochemistry

Few metals are found pure in nature. Most exist in ores, compounds rich in metals that have to be chemically altered to remove the pure metal.

Iron smelting

The most common iron ores are oxides (in which iron is bonded to oxygen), such as hematite (Fe_2O_3). The ores have their oxygen removed in a process called smelting, which takes place in a blast furnace. The reducing agent (the substance that removes the oxygen) is carbon monoxide, a gas that is formed by burning coke, a form of coal. The heat from the combustion of coke also powers the various reactions taking place. Impurities such as silicon dioxide are also removed in the process.



1. $2\text{C} + \text{O}_2 \rightarrow \text{CO}_2 + \text{C} \rightarrow 2\text{CO}$

The coke is more or less pure carbon. It burns near the bottom of the furnace with oxygen, forming carbon dioxide. The carbon dioxide then reacts with more carbon to form carbon monoxide (CO).

2. $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 3\text{CO}_2 + 2\text{Fe}$

The carbon monoxide rises and reacts with the hot ore in the middle of the furnace. Because the carbon in the gas is more reactive than iron, it takes the oxygen ions from the ore, forming pure iron and carbon dioxide gas.

3. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

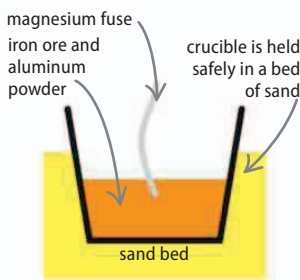
Calcium carbonate (limestone) is also added to the furnace. The heat from the combustion at the bottom makes the calcium carbonate decompose into calcium oxide (quicklime) and carbon dioxide.

4. $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

The quicklime moves to the bottom, where molten iron is being formed. Quicklime is very reactive and reacts with impurities in the iron, such as silicon dioxide, removing them to form a waste product called slag.

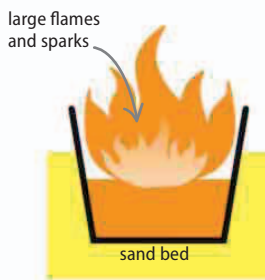
Thermite process

Another way to extract pure iron from its ores is to burn it with pure aluminum, a more reactive metal. This very rapid reaction is called the thermite process and it is exothermic (see page 134). Aluminum is more reactive than iron so it snatches the oxygen from the iron ore, leaving free elemental iron and aluminum oxide.



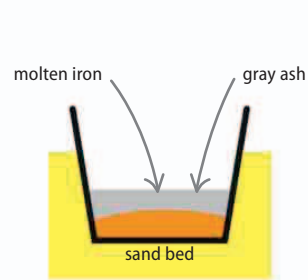
△ Before

Powdered iron ore and aluminum are mixed in a heat-resistant crucible. The reaction is ignited with a strip of magnesium that burns white hot.



△ Reaction

The aluminum snatches the oxygen from the iron, forming aluminum oxide. The reaction releases a large amount of heat with sparks and flames.



△ After

The heat melts the iron and it sinks to the bottom of the crucible. The molten iron is surrounded by the gray crystals of aluminum oxide.